

FREIGHTIQ: BALTIC DRY INDEX INTELLIGENCE NETWORK

Institutional Maritime Analytics, Ensemble Forecasting & Live Data Auto-Update Operational Guide

Target Audience: Dry Bulk Charterers, Vessel Operators, Traders	System Baseline: September 2026 (3,164 Sessions)
Coverage: 2014–2026 Historical Baltic Settlements (10+ Yrs)	Architecture: Super Learner Meta-Ensemble + FastAPI Desk

1. Executive Overview: What is FreightIQ?

FreightIQ is an institutional-grade commercial freight intelligence and decision-support platform designed for dry bulk charterers, vessel operators, commodity trading desks (coking coal, thermal coal, iron ore, bauxite, grain), and maritime supply chain directors. It bridges high-frequency quantitative market forecasting with operational terminal navigation constraints.

The platform addresses four fundamental commercial chartering pillars:

- **Optimal Market Entry & Fixture Timing:** Identifies cyclical rate troughs across Capesize, Panamax, Supramax, and Handysize segments. Recommends prompt vs. deferred strategic entry windows with quantified dollar savings per cargo.
- **Port Infrastructure & Vessel Matching (India East Coast):** Enforces physical draft and LOA constraints across all major East Coast Indian bulk terminals (Dhamra, Gangavaram, Krishnapatnam, Visakhapatnam, Paradip, Ennore, Chennai, Haldia) to eliminate draft exceedance risks.
- **Voyage Economics & Cost Optimization:** Calculates net freight cost per metric ton (\$/MT), sea transit days, port turnaround stay (TPD), VLSFO bunker consumption (@ \$615/MT), port dues, and demurrage exposure.
- **Idle Fleet & Backhaul Triangulation:** Formulates coastal backhaul routes post-discharge to eliminate empty deadhead ballasting (e.g. coastal coal cabotage to Ennore, agricultural export from Kakinada).
- **Early Warning & Meteorological Risk Matrix:** Tracks market variance spikes, South-West Monsoon swell, North-East post-monsoon cyclone windows (Oct–Dec), and berth queues to enforce protective WIBON/WIPON charter terms.

2. Operational Runbook: How to Run FreightIQ

The system is engineered as a lightweight, high-performance stack with zero heavy dependencies.

A. Launching the Interactive Web Workstation (Browser UI)

Execute the FastAPI workstation server from the project directory:

```
cd freightrates
python app.py
```

Open your web browser and navigate to **http://127.0.0.1:8000**. The interactive terminal features real-time dynamic forecasts, interactive horizon toggles (30d, 60d, 90d, 180d), voyage simulators, terminal draft validation, and live CSV export.

B. Running via Command-Line Interface (CLI)

To generate fixture recommendations directly from the terminal:

```
python predict.py --commodity "Coking Coal" --volume 75000 --origin "Hay Point / Dalrymple, Australia"
--dest "Visakhapatnam" --days 30
```

C. Automated Walk-Forward Training & Validation Benchmark

To retrain all four models on the full 10-year Baltic dataset and evaluate out-of-sample metrics:

```
python train.py
```

Detailed validation metrics and 30-day forecast series are written to models/model_report.json.

D. Automated Daily Background Scheduling

- **Windows Task Scheduler:** Double-click setup_windows_task.bat to register a silent Windows Scheduled Task that executes daily at 18:30 (post-London settlement). Logs are written to logs/scrapper.log.
- **Python Daemon:** Run python bdi_live_scraper.py --schedule 18:30 for continuous background execution.

3. Quantitative Forecasting Models & 10-Year Benchmark

FreightIQ utilizes a **Super Learner Stacking Meta-Ensemble** trained and validated via walk-forward out-of-sample testing across full market cycles (the 2016 commodity slump, 2020–2022 pandemic disruption, Red Sea canal diversions, and 2024–2026 recovery).

Model Architecture	Out-of-Sample MAE	RMSE	MAPE (%)	Directional Acc.	R² Score
Gradient Boosted Trees (GBDT)	\$34.78	\$47.56	1.83%	70.85%	0.9930
Deep Neural Network (MLP-ResNet)	\$36.31	\$49.53	1.90%	71.18%	0.9925
Ridge Autoregressive AR(60)	\$35.50	\$48.87	1.86%	69.89%	0.9927
Super Learner Meta-Ensemble	\$36.82	\$50.59	1.91%	69.40%	0.9921

34-Dimensional Technical & Maritime Feature Matrix:

- **Trend & Moving Averages:** SMA 5, 10, 20, 50, 100, 200, Golden/Death Cross indicator, EMA 12, 26, MACD, Signal, Histogram.
- **Momentum & Mean Reversion:** RSI-14, Momentum Rate of Change (ROC 5, 20, 60).
- **Multi-Scale Volatility:** Rolling Std (5, 20, 60d), Parkinson High-Low Volatility, Bollinger Bands (%B and Bandwidth).
- **Cyclical Harmonics:** Multi-frequency Fourier sine/cosine series (Annual 365.25d, Semi-Annual 182.6d, Quarterly 91.3d).
- **Autoregressive Lags:** Historical Return Lags 1, 2, 3, 5, 10, 20, 30, and 60 trading sessions.

4. Live BDI Data Auto-Update Engine (Web Scraping)

To ensure models never go stale, FreightIQ features an institutional scraping and automated model retraining pipeline (bdi_live_scraper.py).

A. Multi-Source Scraping Hierarchy & Failover

- **Primary (TradingEconomics):** Direct JSON script extraction via TEChartsMeta (immune to HTML DOM table column shifting), plus dynamic <th> column header mapping.
- **Secondary (Seaandjob News Wire):** Scrapes official daily Baltic Exchange market settlement reports (e.g. 'The Baltic dry index fell 1.8% to 3445 on Monday').
- **Tertiary (Hellenic Shipping News):** Parses daily fixture indices across Capesize, Panamax, and Supramax.
- **Quaternary (Investing.com):** Browser session emulation with realistic user headers.

B. Institutional Validation Gate (Circuit-Breakers)

Candidate settlements must pass 5 strict validation checks before writing to CSV:

1. **Reasonable Range:** Index price must be bounded between \$200 and \$25,000.
2. **Weekend Circuit-Breaker:** Rejects Saturday or Sunday prints (Mon–Fri trading sessions only).
3. **15% Daily Movement Circuit-Breaker:** Rejects prints moving > 15% from prior close to prevent misparsed column anomalies.
4. **Chronology Check:** Reject backdated or stale prints (candidate date ≥ latest recorded CSV date).
5. **Internal Consistency:** Verified that Price - Prior Close == Reported Point Change within ±5 points.

C. Canonical CSV Sync & Automated Model Retraining

- **Deterministic Path Resolution:** The canonical CSV in freightrates/ is updated directly, and root CSV synchronized.
- **Ordered Deduplication:** Checks if date exists. If new, inserts in descending chronological order.
- **Automatic Retraining:** Retrains GBDT, MLP-ResNet, Ridge, and Meta-Ensemble, writing updated weights and forecasts to models/model_report.json in ~15 seconds.
- **In-App Web 1-Click Sync:** Clicking **[LIVE AUTO-SYNC]** on the web dashboard scrapes fresh data, retrains models, and hot-reloads data in memory with cache-busting so the browser immediately reflects the fresh settlement.

5. End-to-End System Architecture: How It All Works

Stage	Component / File	Key Operation & Output
1. Live Ingestion	bdi_live_scraper.py	Scrapes live BDI settlement, runs validation gate, appends to CSV with deduplication.
2. Feature Eng.	baltic_data.py	Constructs 34-dim technical matrix (SMAs, MACD, RSI, volatility, Fourier harmonics, lags).
3. Model Training	forecasting_models.py train.py	Walk-forward cross validation across GBDT, MLP, Ridge. Fits Meta-Ensemble and saves report.
4. Freight Engine	freight_engine.py maritime_data.py	Evaluates voyage economics, terminal draft limits, bunker costs, demurrage, and laycan timing.
5. User Interface	app.py (FastAPI + Web UI)	Serves interactive workstation at port 8000. Real-time forecasting, simulation, and 1-click sync.

Summary & Next Steps: FreightIQ is fully configured, validated on 3,164 daily market settlements, and synced to GitHub (himamshu07-dot/FREIGHTIQ). You can launch the server using python app.py or run daily updates via setup_windows_task.bat.